

Digital Forensics & Incident Response

UNIT 4 - Network Forensics & Log Analysis

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Introduction to Network Forensics



Introduction to Network Forensics

What is Network Forensics?

Network forensics is the process of capturing, examining and analysing network traffic and related network data to investigate security incidents and reconstruct network activity.

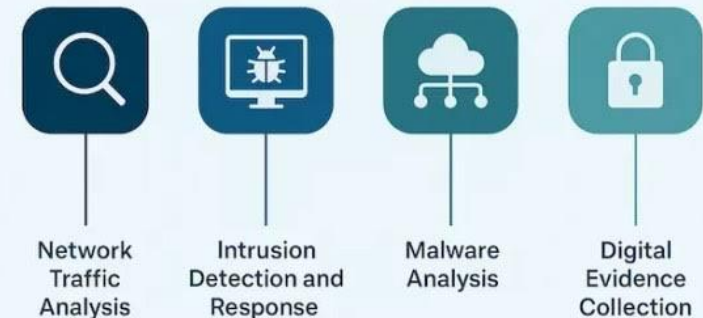
When devices communicate over a network, they generate digital traces such as packets, connections, DNS requests, firewall records and proxy logs. These traces can help investigators understand **who communicated with whom, when the communication occurred, what protocols were used and whether the activity was suspicious.**

Importance of Network Forensics

Network forensics is useful for investigating:

- Cyberattacks and intrusions
- Unauthorised access
- Data exfiltration
- Malware communication
- Suspicious network connections
- Command-and-control activity

Types of Network Forensics



Understanding Network Communication

Before analysing network evidence, it is important to understand how devices communicate.

A **network** allows computers, servers, mobile devices and other systems to exchange information. This communication is divided into structured units called **packets**, which contain information about the communication.

A network communication may contain:

Source: Device initiating the communication

Destination: Device receiving the communication

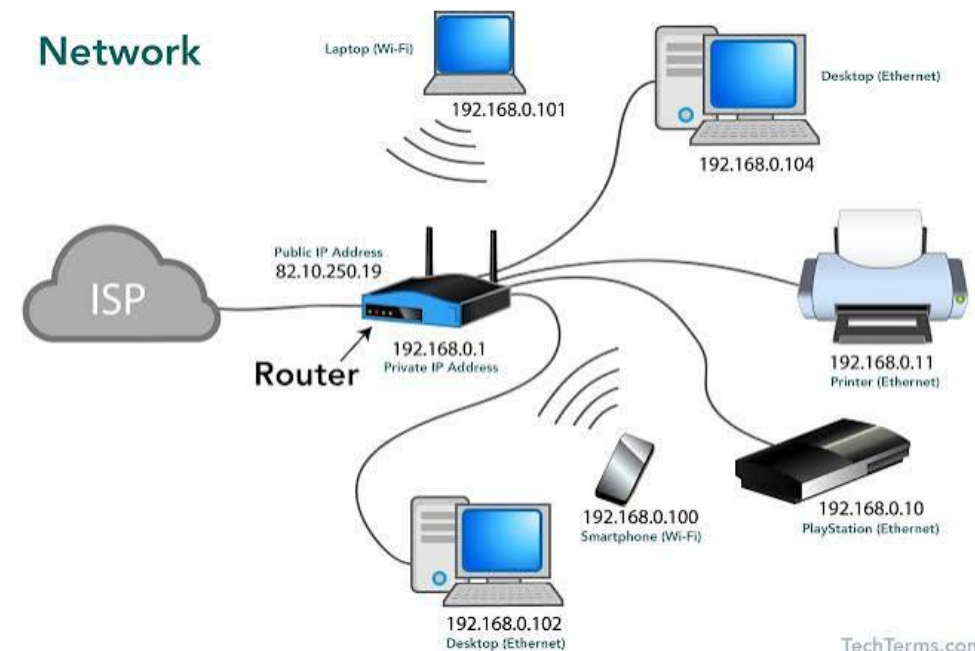
Protocol: Rules used for communication

Port: Identifies a network service or application

Timestamp: Indicates when communication occurred

Forensic Importance

By analysing these characteristics, investigators can reconstruct communication patterns and identify unusual activity.



Network Protocols and Ports

What is a Network Protocol?

A **network protocol** is a defined set of rules that determines how devices communicate and exchange data.

Common protocols encountered during forensic investigations include:

- **TCP** - Reliable, connection-oriented communication
- **UDP** - Connectionless communication
- **HTTP/HTTPS** - Web communication
- **DNS** - Domain-name resolution
- **FTP** - File transfer

SMTP - Email transmission

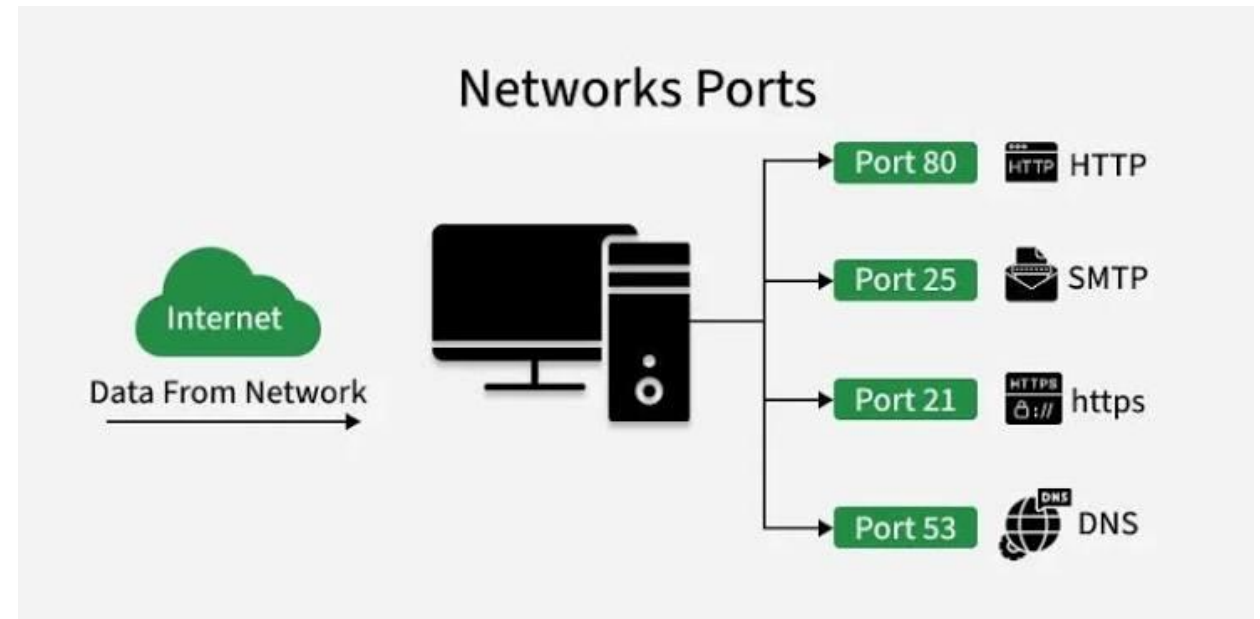
What is a Port?

A **port number** identifies a logical communication endpoint associated with a network service.

For example:

- **TCP 80** → HTTP
- **TCP 443** → HTTPS
- **TCP 21** → FTP
- **TCP 25** → SMTP

Understanding protocols and ports helps investigators interpret captured network traffic.



Network Layer vs Endpoint Layer Evidence

Network and endpoint evidence provide different perspectives of the same incident.

Network Layer Evidence	Endpoint Layer Evidence
Shows communication between systems .	Shows activity occurring on an individual system .
Examples: • IP addresses • Ports • Protocols • DNS requests • Network connections • Packet contents	Examples: • Processes • Files • User activity • Event logs • Registry artefacts

Investigation Concept
Network evidence tells us about communication.
Endpoint evidence tells us about activity on the communicating system.
Combining both provides a stronger understanding of an incident.



Network Forensics Process

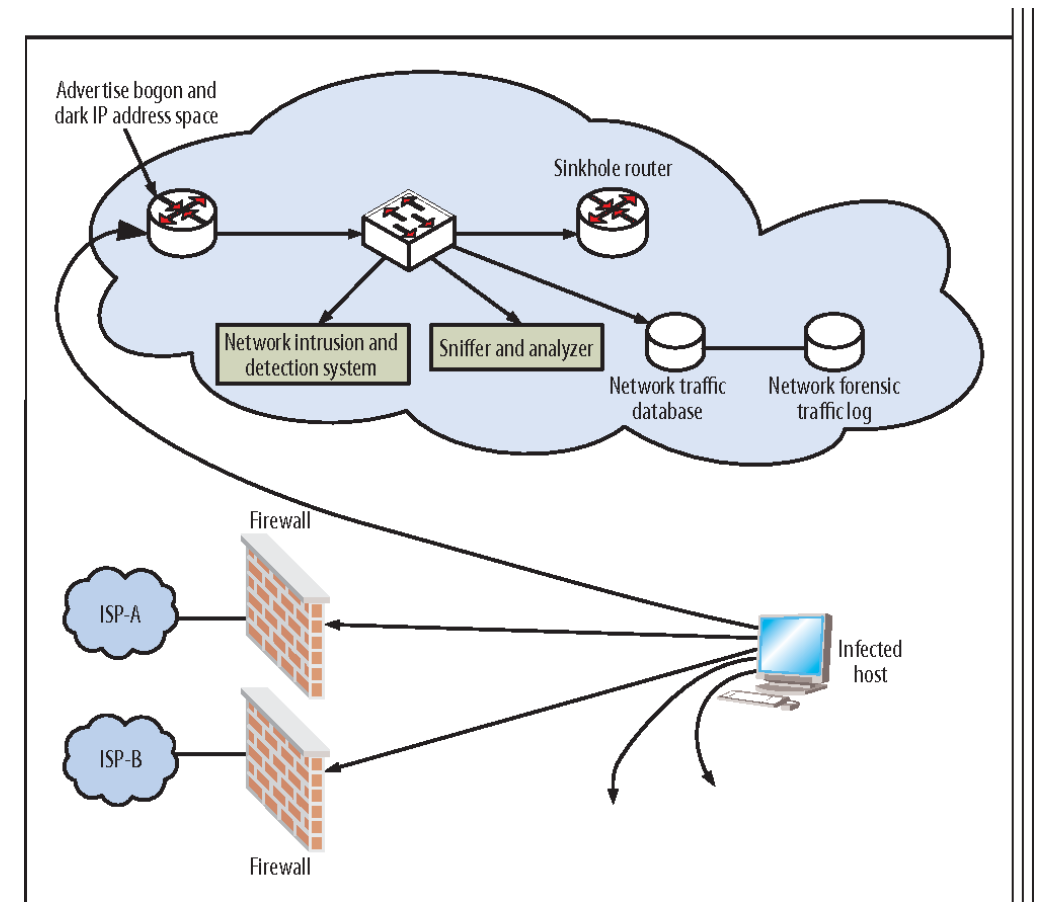
Network Forensic Investigation

A network forensic investigation follows a structured process.

**Identify Evidence Sources → Collect Network Data →
Preserve Evidence → Examine Traffic and Logs →
Identify Suspicious Activity → Correlate Network & Endpoint Evidence
→ Reconstruct Events → Document Findings**

Objective

The objective is to transform raw network information into an understandable reconstruction of what occurred during the incident.



Introduction to Wireshark

What is Wireshark?

Wireshark is a network protocol analyser used to capture and examine network packets. It provides detailed visibility into network communications and allows investigators to inspect individual packets, protocols, addresses, ports and other network information.

Wireshark is useful for:

- Packet analysis
- Protocol investigation
- Identifying suspicious traffic
- Following network conversations
- Extracting transferred objects
- Analysing DNS and HTTP traffic
- Reconstructing network timelines



Wireshark Display Filters

Large packet captures can contain thousands or millions of packets. **Display filters** allow investigators to focus on traffic relevant to their investigation.

Common Filters

ip.addr: Filters traffic associated with an IP address.

tcp.port: Filters traffic associated with a TCP port.

http: Displays HTTP traffic.

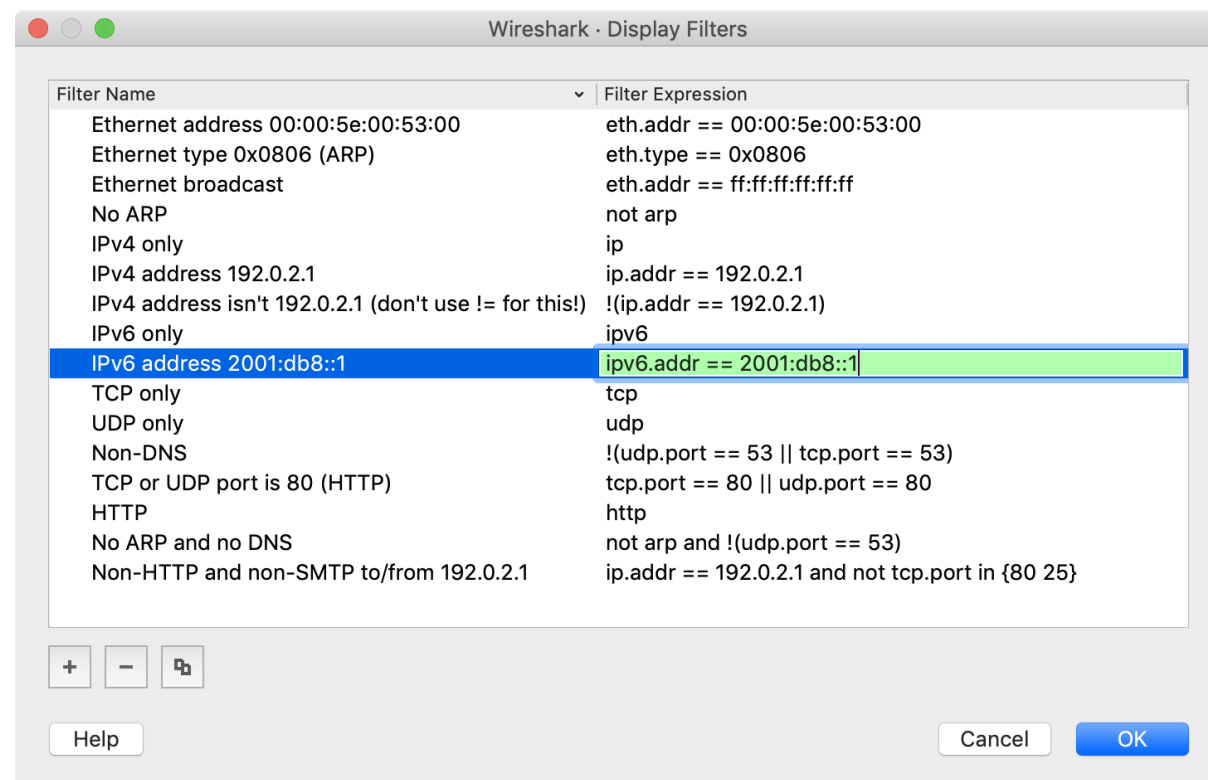
ftp: Displays FTP traffic.

Dns: Displays DNS traffic.

smtp: Displays SMTP traffic.

Purpose

Filtering reduces unnecessary information and allows investigators to focus on specific communication patterns.



Following TCP Streams

What is a TCP Stream?

A **TCP stream** represents the continuous communication between two endpoints over a TCP connection. Instead of examining individual packets separately, investigators can reconstruct the communication as a continuous conversation.

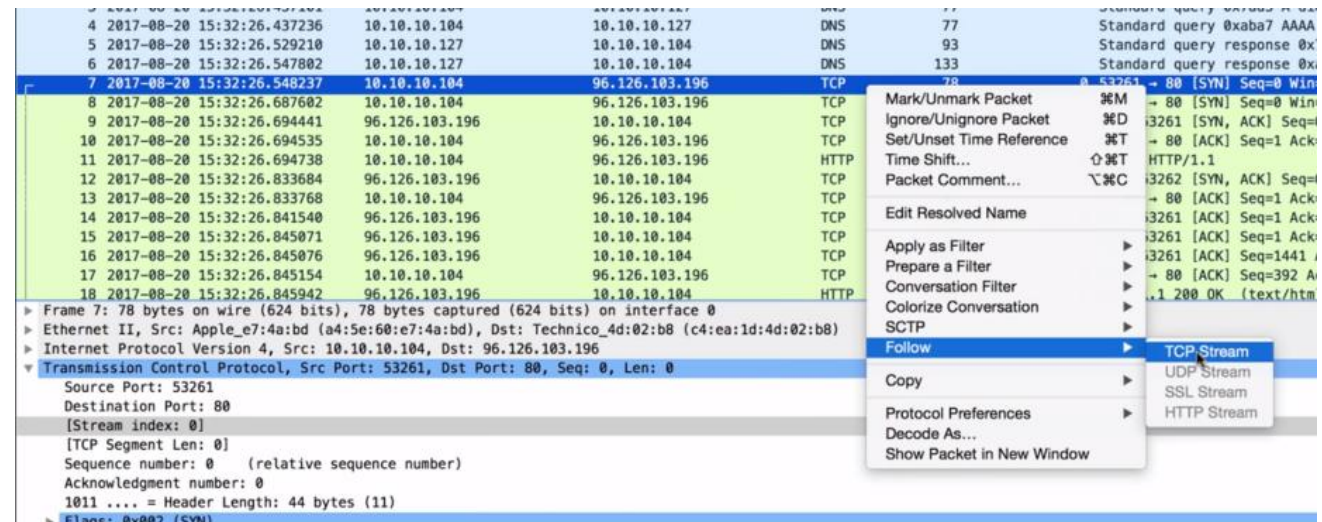
Why is this useful?

Following a TCP stream can help investigators understand:

- What was communicated
- Which systems participated
- The sequence of communication
- Whether files or credentials were transmitted
- Whether the communication appears suspicious

Concept

Individual Packets → Reconstructed Conversation
→ Understand Communication



Extracting Transferred Files

Network traffic may contain files transferred between systems.
Wireshark can provide mechanisms for extracting objects from supported protocols.

Common Investigation Path

File → Export Objects

The investigator can examine supported protocols such as:

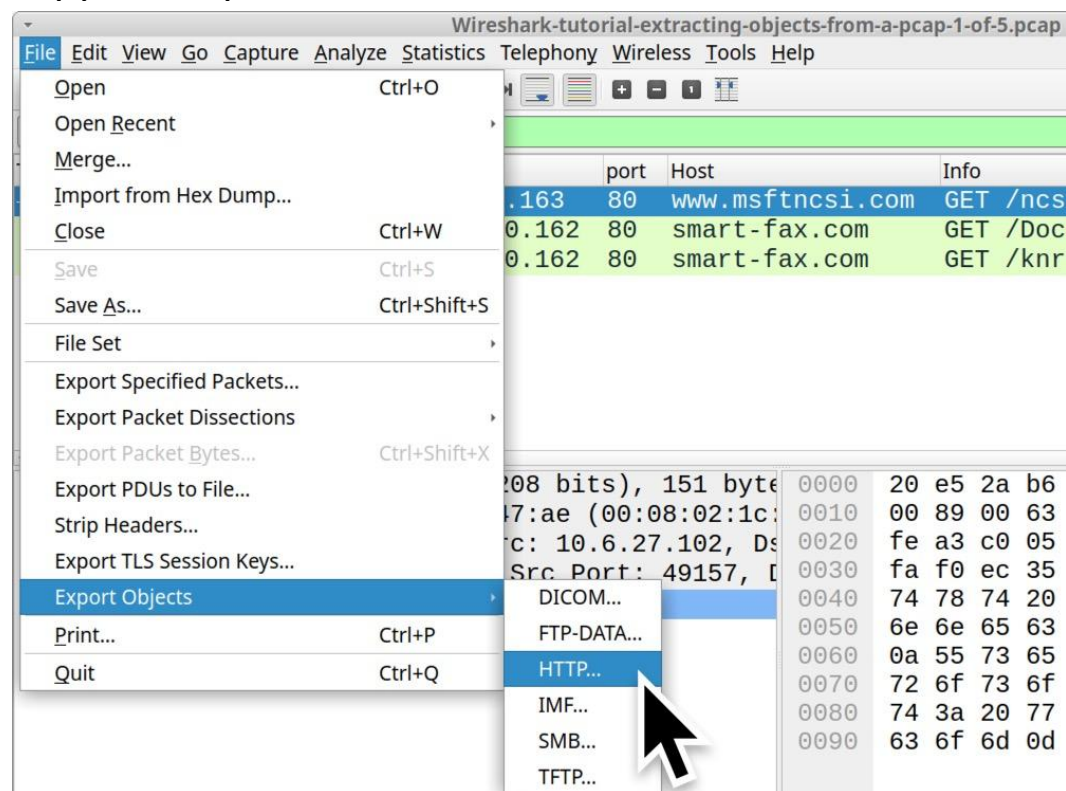
- HTTP
- SMB

Forensic Value

Extracted files can be examined separately for:

- File type
- Metadata
- Hash values
- Suspicious content
- Malware indicators

This creates a connection between **network evidence** and **file-system evidence**.



Cleartext Credentials in Network Traffic

What are Cleartext Credentials?

Cleartext credentials are usernames, passwords or authentication information transmitted without adequate encryption. Protocols such as traditional **FTP** and **HTTP** can expose sensitive information when credentials are transmitted without encryption.

Forensic Investigation

Investigators can examine captured traffic for:

- Usernames
- Passwords
- Authentication requests
- Session information

Security Significance

If credentials are visible in network traffic, an attacker monitoring that traffic may potentially obtain them.

OST

Source	Destination	Pro
172.99.96.253	160.153.129.234	HT

RT: http://www.sababank.com/signin.php]

/1]

ame: 1129]

ytes

ded: application/x-www-form-urlencoded

rname" = "Ibrahim_Diyeb"

sword" = "yemen_123"

n" = "signin"

0d 0a 43 6f 6e 74 65 6e 74 2d 4c

3a 20 35 33 0d 0a 43 6f 6f 6b 69

50 53 45 53 53 49 44 3d 34 31 32

30 63 35 36 37 34 35 61 63 66 34

39 36 34 63 32 62 65 35 3b 20 6c

72 61 62 69 63 0d 0a 43 6f 6e 6e

6e 3a 20 6b 65 65 70 2d 61 6c 69

70 67 72 61 64 65 2d 49 6e 73 65

52 65 71 75 65 73 74 73 3a 20 31

73 65 72 6e 61 6d 65 3d 49 62 72

44 69 79 65 62 26 70 61 73 73 77

coded..Content-L

ength: 53..Cooki

e: PHPSESSID=412

354120c5 6745acf4

1b8e2964 c2be5; 1

ang=arabic..Conn

ection: keep-ali

ve..Upgrade-Inse

cure-Requests: 1

....user name=Ibr

ahim_Diy eb&passw

This is one reason secure protocols such as **HTTPS** and **encrypted authentication mechanisms** are important.



Command-and-Control (C2) Beacons

What is C2?

Command and Control (C2) refers to communication between compromised systems and infrastructure controlled by an attacker.

Malware may periodically contact a remote server to:

- Receive commands

- Send system information

- Download additional payloads

- Report the status of the infected system

Beaconing Pattern

One suspicious pattern is **regular communication with the same destination at repeated intervals**.

For example:

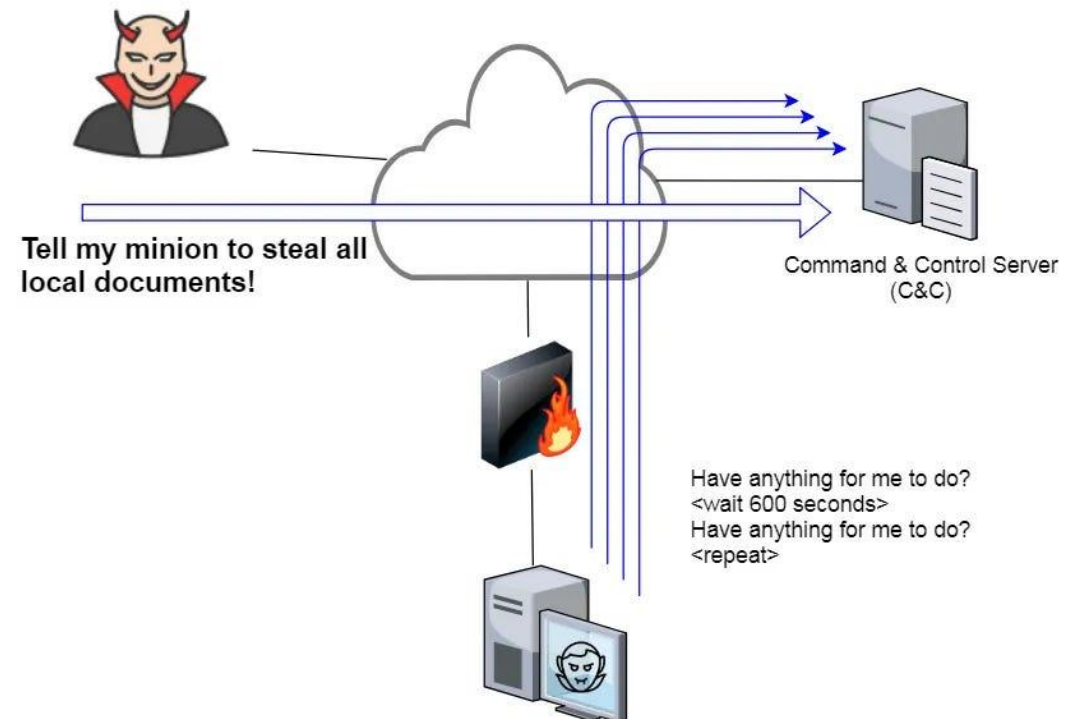
10:00 → IP A

10:05 → IP A

10:10 → IP A

10:15 → IP A

Regular periodic communication may indicate beaconing and should be investigated further.



DNS Tunnelling

What is DNS Tunnelling?

DNS tunnelling is a technique in which DNS queries or responses are abused to carry information through DNS traffic. DNS is normally used to translate domain names into IP addresses. Because DNS traffic is commonly permitted through networks, attackers may attempt to misuse it for communication or data transfer.

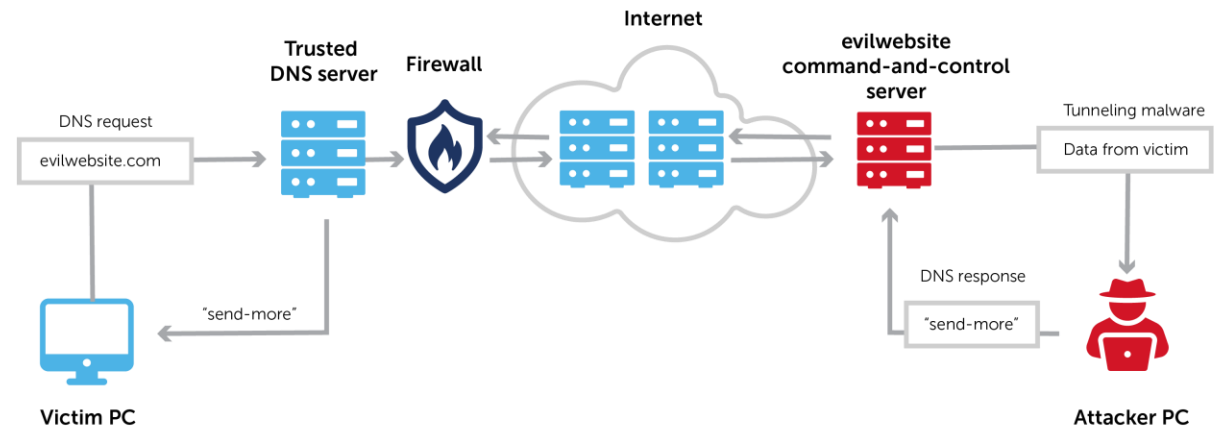
Possible Indicators

- Unusually long subdomain names
- Large numbers of DNS queries
- High-entropy or random-looking subdomains
- Repeated queries to the same unusual domain
- Abnormally large amounts of DNS traffic

Forensic Objective

Investigators analyse DNS patterns to determine whether normal DNS activity has potentially been abused.

DNS tunneling



Network Timeline Reconstruction

What is a Network Timeline?

A network timeline arranges network events according to their timestamps to reconstruct the sequence of an incident.

Example

10:02 — DNS query to suspicious domain

10:03 — Connection established with remote IP

10:04 — Data transferred

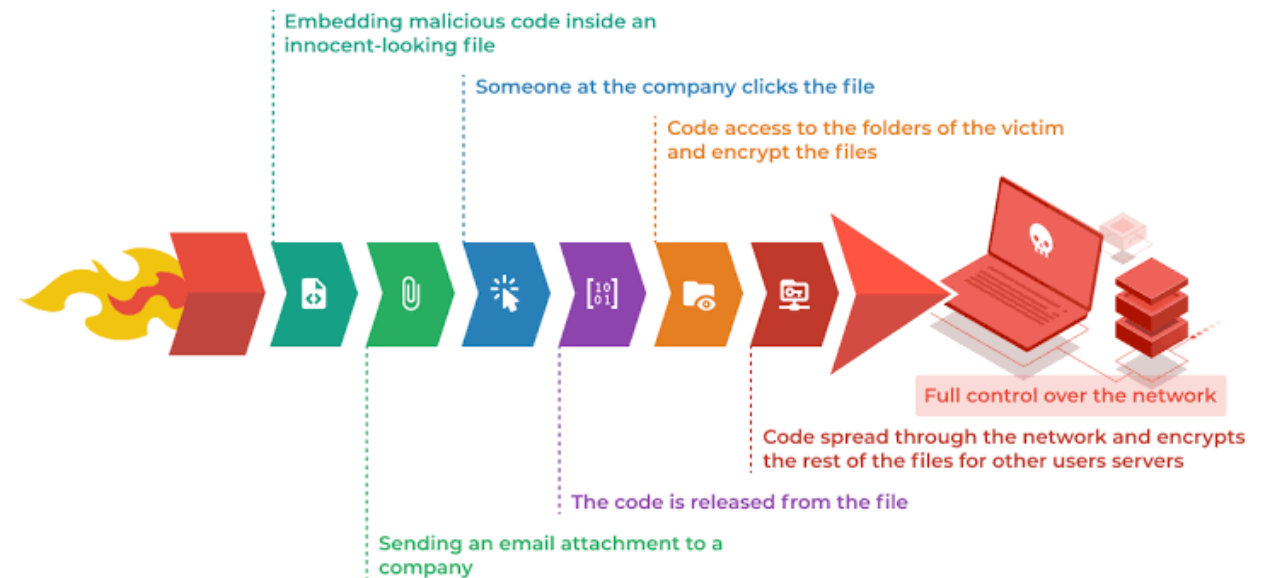
10:05 — Connection terminated

10:10 — Same domain queried again

Importance

Timeline analysis helps investigators understand:

- What happened first
- What happened next
- How long activity lasted
- Which events occurred together
- Whether different evidence sources support the same sequence



NetworkMiner and Passive Network Analysis



NetworkMiner

What is NetworkMiner?

NetworkMiner is a passive network-analysis tool that can extract information from captured network traffic. Instead of focusing only on individual packets, it helps investigators identify **hosts, transferred files, credentials, images and other network artefacts**.

Major Capabilities

- Host identification
- OS fingerprinting
- File extraction
- Image extraction
- Credential extraction
- X.509 certificate extraction



NetworkMiner can therefore provide a more organised view of information contained within a PCAP file.



PCAP and Passive Network Analysis

What is PCAP?

PCAP is a format commonly used to store captured network packets.

A PCAP file can contain a record of network communication captured during monitoring or investigation.

Passive Analysis

Passive network analysis examines already captured traffic without actively interacting with the systems being investigated.

Investigation Flow

PCAP File → Load into NetworkMiner → Identify Hosts
→ Extract Artefacts → Analyse Communications →
→ Correlate with Other Evidence

PCAP File Structure



Email Forensics and Email Authentication



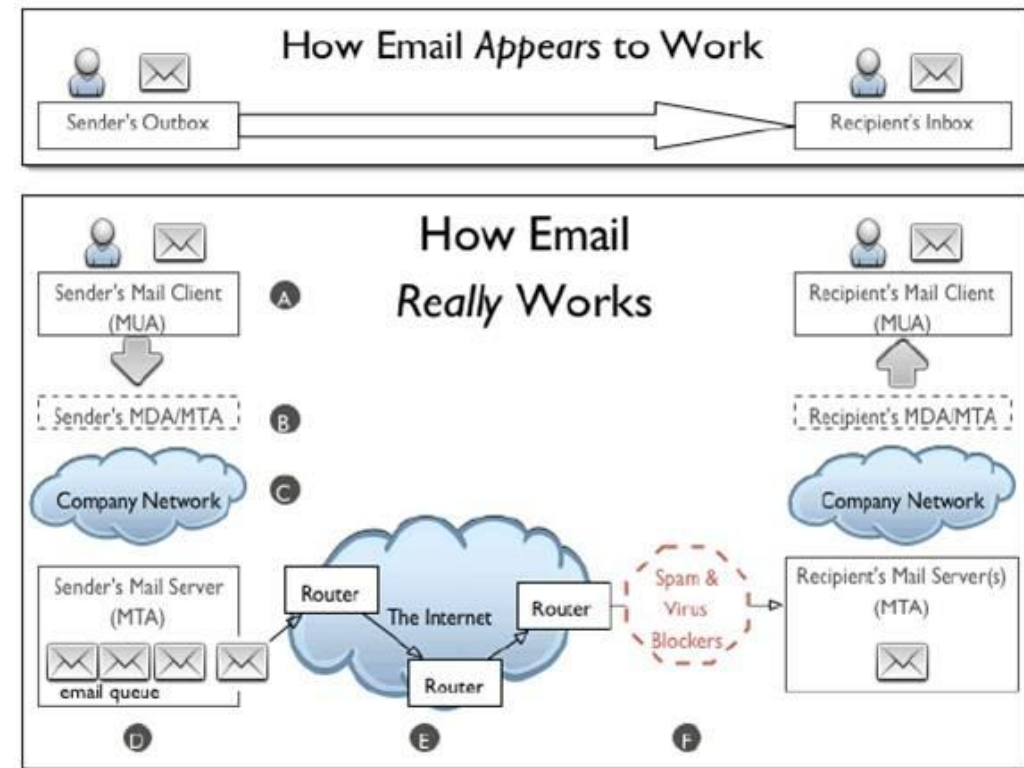
Introduction to Email Forensics

Email forensics is the examination of email content, metadata and transmission information to determine the origin, path and authenticity of an email.

An email contains more information than what is visible in the message body.

Investigators may examine:

- Email headers
- Sender and recipient information
- Received headers
- IP addresses
- Timestamps
- Authentication results
- Attachments
- Message content



Objective: Determine **where** an email came from, **how** it travelled and **whether** its identity may have been manipulated. 

Reading Raw Email Headers

What are Email Headers?

Email headers contain technical information describing how an email was processed and delivered. Important fields include:

From: Displayed sender address

To: Recipient

Date: Message timestamp

Subject: Message subject

Received: Server-by-server delivery information

Message-ID: Identifier associated with the message

Forensic Importance

Raw headers can reveal information that is not visible in the normal email interface.



Tracing an Email Through Received Headers

Received Header

Every mail server that processes an email can add a **Received** header.

By examining these headers, investigators can reconstruct the route taken by the message.

Simplified Flow

Originating System



Mail Server 1



Mail Server 2



Recipient Mail Server



Recipient

Folders
Last Refresh:
Mon, 1:15 pm
(Check mail)

- INBOX (0/1)
- Drafts
- Sent
- Trash (Purge)
- Archives
- CS654
- EE699
- Junk
- Mtech
- Folder Sizes

Current Folder: **INBOX** [Sign Out](#)
[Help \(LAN\)](#)

[Compose](#) [Addresses](#) [Folders](#) [Options](#) [Search](#) [Help](#) [Calendar](#) [Auto Response](#)

Viewing Full Header - [View message](#)

Return-Path: <hacker@iitk.ac.in> 6

X-Original-To: soum@iitk.ac.in

Delivered-To: soum@iitk.ac.in 7

Received: from mail2.iitk.ac.in (mail2.iitk.ac.in [172.31.1.214])
by newmailhost.cc.iitk.ac.in (Postfix) with ESMTPS id 9F6F91A000206
for <soum@iitk.ac.in>; Fri, 18 Sep 2020 20:45:54 +0530 (IST)

Received: from smtp.cc.iitk.ac.in (smtp.cc.iitk.ac.in [172.31.1.22])
by mail2.iitk.ac.in (Postfix) with ESMTP id 93B9E1800088
for <soum@iitk.ac.in>; Fri, 18 Sep 2020 20:45:54 +0530 (IST) 2

Received: from unknown (unknown [172.25.14.111]) 3
(using TLSv1.2 with cipher ECDHE-RSA-AES256-GCM-SHA384 (256/256 bits))
(No client certificate requested)
(Authenticated sender: mailadm 5)
by smtp.cc.iitk.ac.in (Postfix) with SMTP id 36FA4240022F
for <soum@iitk.ac.in>; Fri, 18 Sep 2020 20:45:06 +0530 (IST) 4

Subject: Special Offer for you

From: Super Man <superman@iitk.ac.in>

To: Some Other User <superuser@iitk.ac.in> 1

Investigators can work backwards through the Received headers to understand the path of the email and identify the earliest relevant originating information.



SPF, DKIM and DMARC

Modern email authentication uses several mechanisms to reduce spoofing and improve trust.

SPF - Sender Policy Framework

SPF allows a domain to specify which mail servers are authorised to send email for that domain.

DKIM - DomainKeys Identified Mail

DKIM uses a cryptographic signature to allow receiving systems to verify that certain parts of the message were associated with the sending domain and were not improperly altered in transit.

DMARC - Domain-based Message Authentication, Reporting & Conformance

DMARC allows a domain owner to specify how receiving systems should handle messages that fail relevant SPF/DKIM alignment checks.

These authentication results can help investigators assess whether an email is potentially **spoofed or impersonated**.



Splunk for Incident Response and Log Analysis



Introduction to Splunk for Incident Response

What is Splunk?

Splunk is a platform used to collect, search, analyse and visualise machine-generated data such as logs and security events. In incident response, Splunk can bring together information from multiple sources and allow investigators to search for patterns across those sources.

Security Investigation Uses

- Authentication analysis
- Firewall-log analysis
- Incident correlation
- Timeline construction
- Detection of suspicious activity
- Investigation dashboards



SPL for Incident Investigation

What is SPL?

Search Processing Language (SPL) is the search language used by Splunk to query and analyse indexed data.

A search can identify patterns across large quantities of log data.

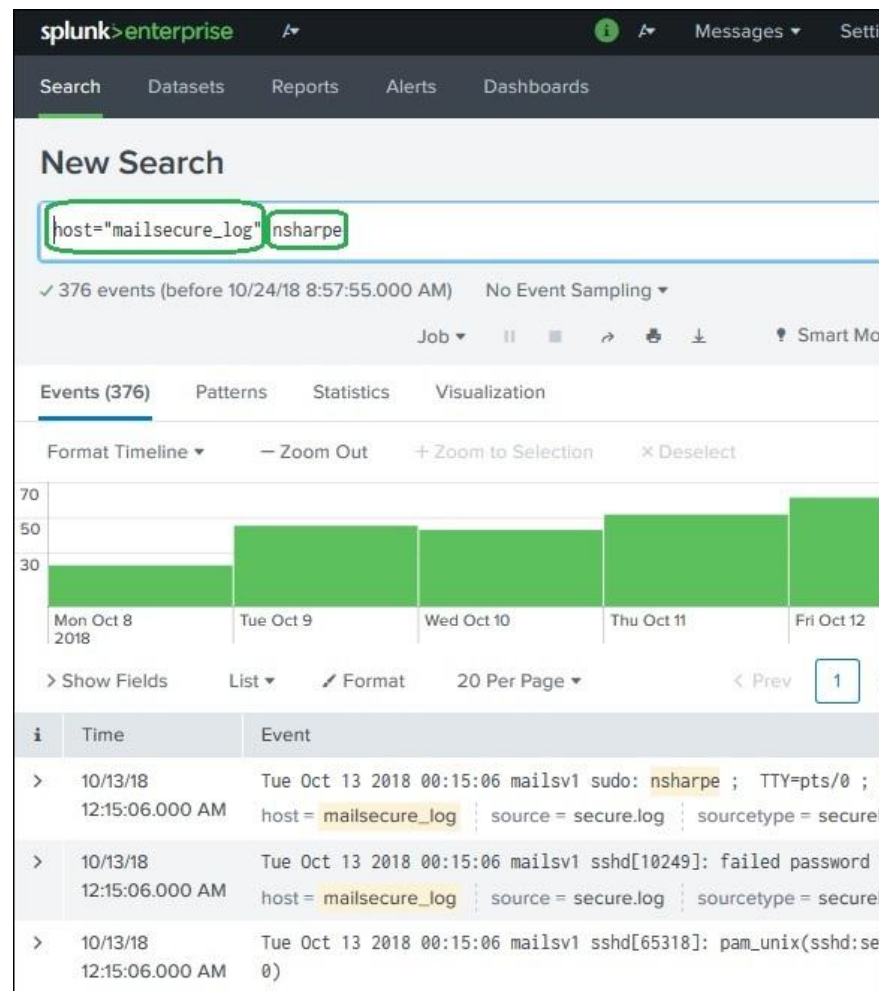
Example

`index=* EventCode=4625 | timechart span=5m count`

This type of search can help visualise **failed logon activity over time**.

Investigation Value

Instead of manually reviewing thousands of log entries, investigators can use SPL to identify patterns, trends and unusual activity.



Correlating Multiple Log Sources

A security incident may generate evidence in multiple systems.

For example:

Windows Security Log

→ Failed login

Firewall Log

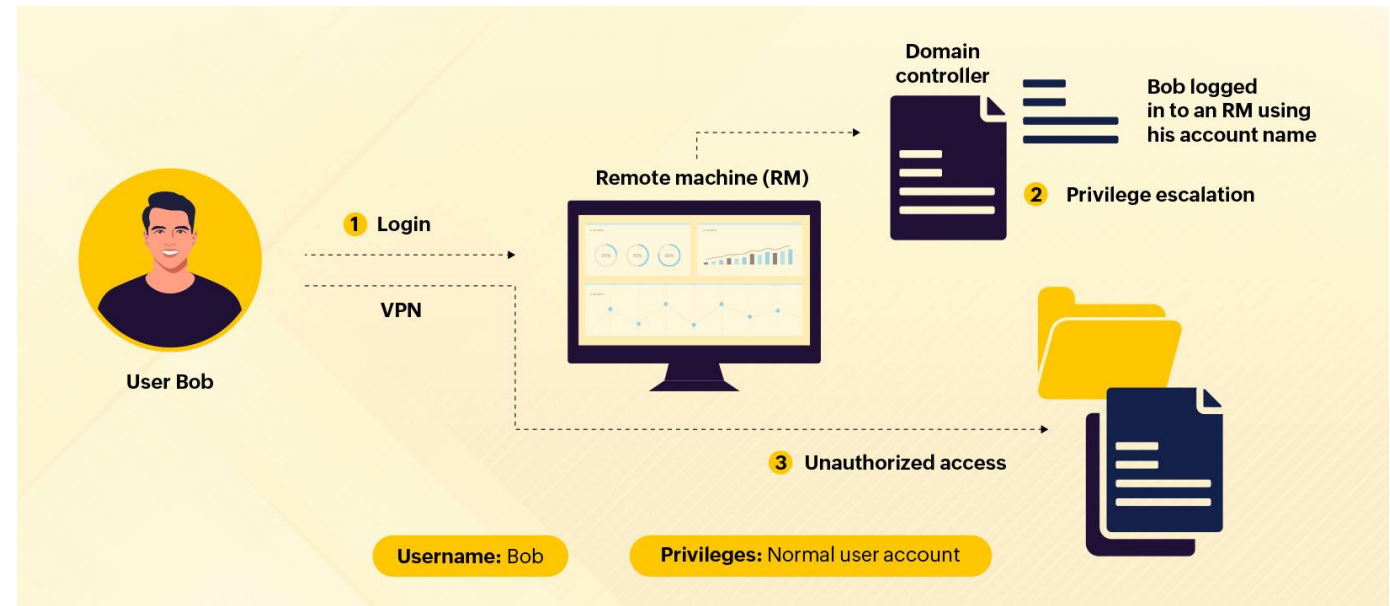
→ Source IP blocked

Endpoint Log

→ Successful login afterwards

Correlation

When these events are connected using common attributes such as **IP address, username and timestamp**, investigators can develop a stronger understanding of the incident.



Key Principle: Correlation transforms individual log events into an investigation timeline.



Detecting Lateral Movement

Lateral movement is the activity of an attacker moving from one compromised system to other systems within a network. After gaining access to one workstation, an attacker may attempt to access servers or additional endpoints.

Forensic Indicators

- Investigators may look for:
- Unusual successful logons
- Remote authentication
- Unexpected source systems
- New destination systems
- Privileged account activity

Windows Event ID **4624** can provide useful information when analysing successful logons.

Investigation Objective

Identify whether authentication activity represents normal administrative behaviour or potentially suspicious movement between systems.

WHAT IS LATERAL MOVEMENT

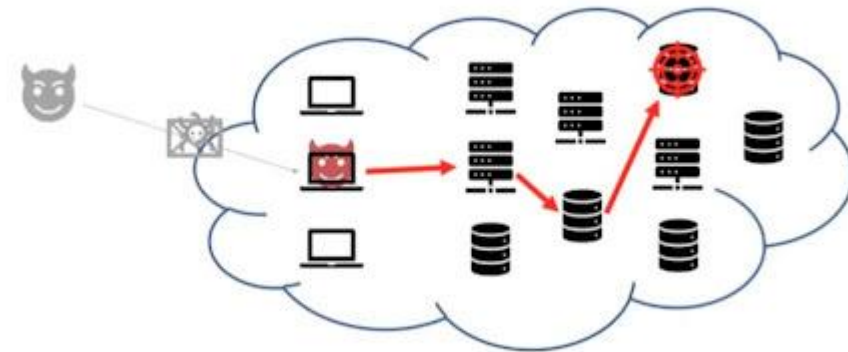


Figure 1: Lateral movement, depicted as red arrows, is the set of attacker movements between *internal* machines in an enterprise.



Splunk Investigation Dashboard

Security Investigation Dashboard

A Splunk dashboard can bring multiple investigation views together.

Possible Components

Failed Login Trend

Shows authentication failures over time.

Source IP Analysis

Shows systems generating suspicious activity.

Firewall Activity

Shows blocked or allowed connections.

Successful Logons

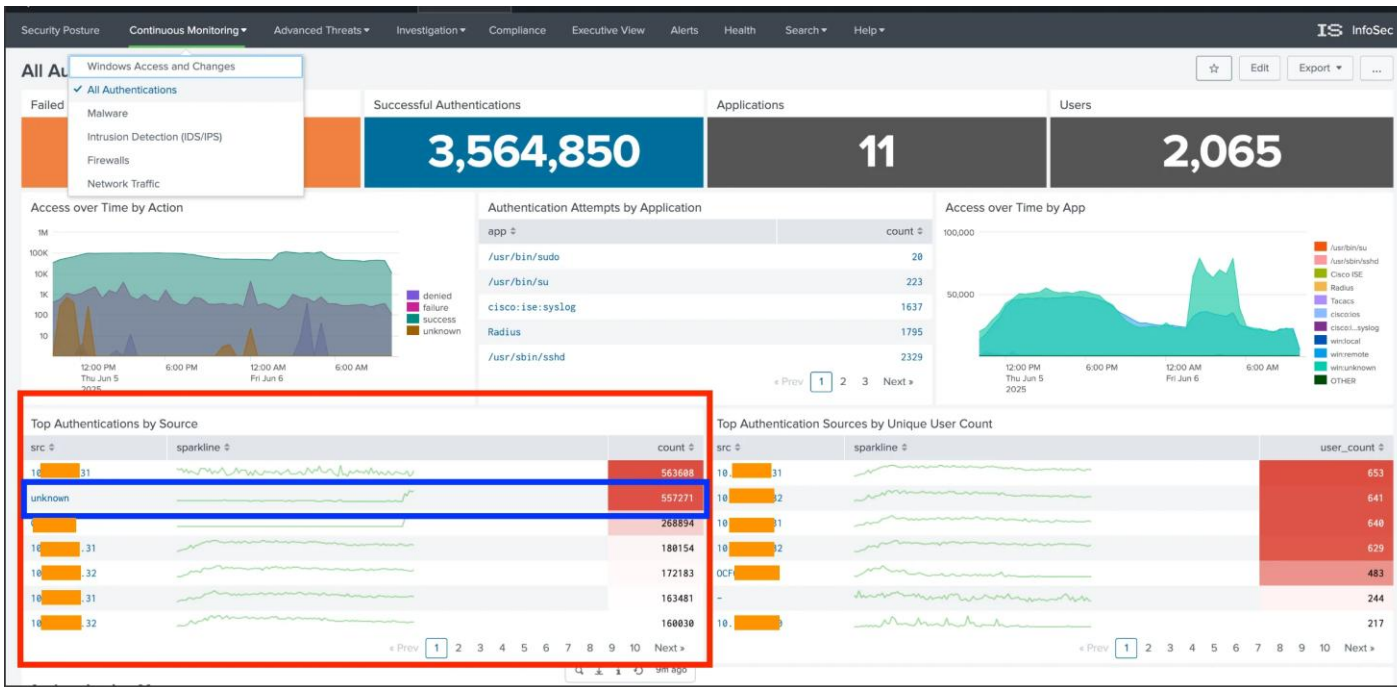
Shows authentication following suspicious activity.

Timeline

Combines related events chronologically.

Benefit

Dashboards provide investigators with a **centralised visual view of an incident.**



Indicators of Compromise (IOCs) & Malware Analysis



Indicators of Compromise (IOCs)

An **Indicator of Compromise (IOC)** is a piece of information that may indicate that a system, network or account has been compromised.

IOCs are observable characteristics associated with potentially malicious activity.

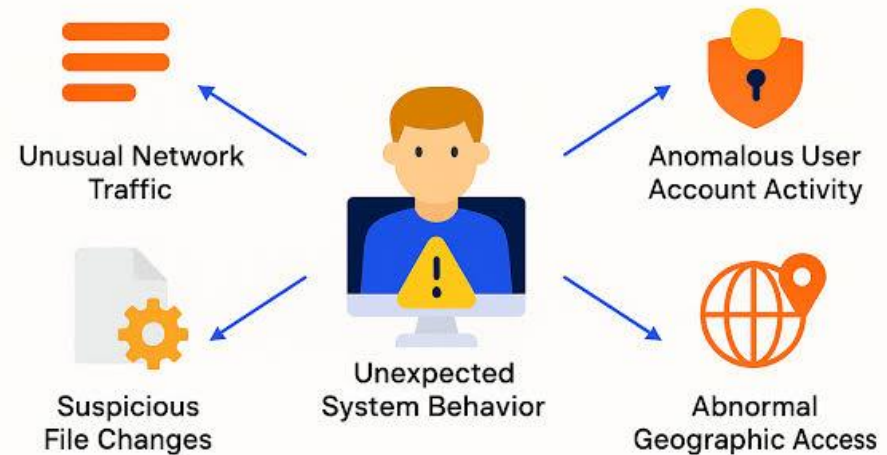
Common IOCs

- IP address
- Domain name
- File hash
- Registry key
- Mutex
- File path

Example

A known malicious file hash can be searched across an organisation's systems to determine whether the same file exists elsewhere.

Common Examples of Indicators of Compromise



IOC Investigation

IOCs can be collected from different sources during an investigation.

Example

A malware investigation identifies:

Malicious IP: 203.0.113.10

Suspicious Domain: example-malicious-domain.com

File Hash: SHA-256 value

File Path: Suspicious executable location

These indicators can then be searched across:

- Endpoint logs
- Network traffic
- SIEM systems
- Forensic images
- Threat-intelligence sources

Objective

Determine whether the same indicators appear elsewhere in the environment.



VirusTotal for Hash Analysis

VirusTotal is a service that aggregates malware and threat intelligence information from multiple security sources. A file's cryptographic hash can be searched to determine whether the hash is already associated with known detections or reports.

Forensic Use

An investigator may use a file hash to:

- Search for known malware
- Review detection results
- Examine associated threat intelligence
- Compare information from multiple security vendors



Important Principle

A VirusTotal result should be treated as **supporting intelligence**, not as the sole basis for a forensic conclusion.



Introduction to Malware Analysis

Malware analysis is the examination of malicious or potentially malicious software to understand its characteristics, behaviour, capabilities and potential impact.

The goal is to determine:

- What the malware is
- How it works
- What it may modify
- How it communicates
- How it persists
- What indicators it creates

Two Broad Approaches

Static Analysis

Examines a file without executing it.

Dynamic Analysis

Examines behaviour while the program is executed in a controlled environment.

This unit introduces **basic static analysis**.



Static Malware Analysis

Static analysis examines a suspicious file without executing it.

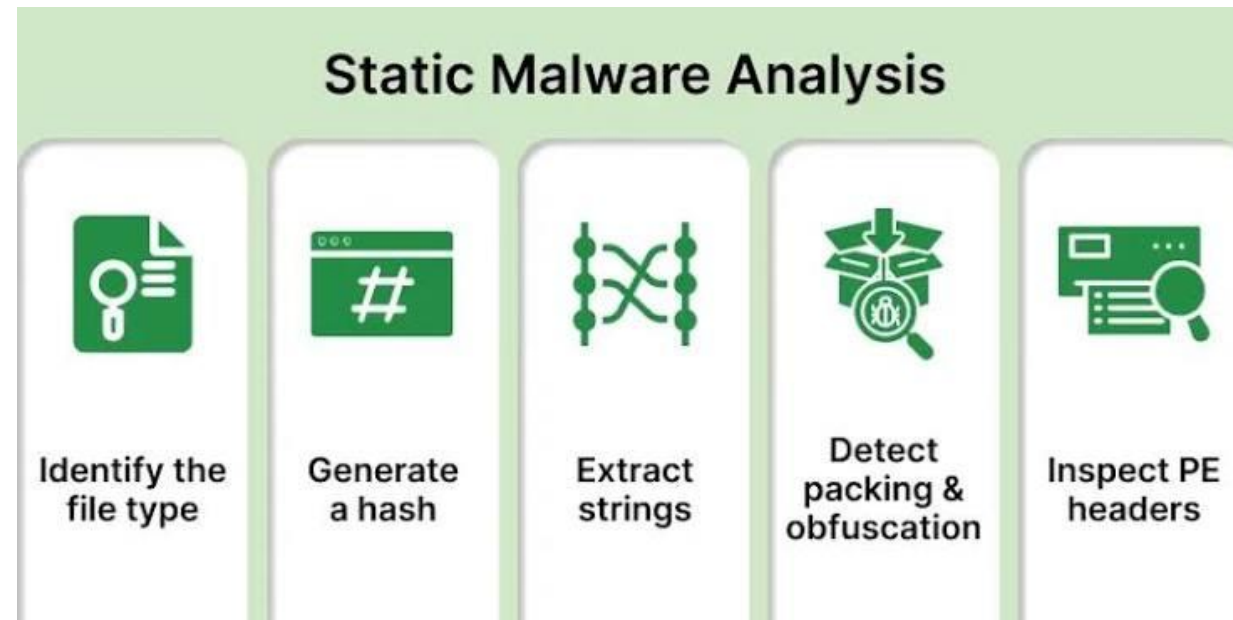
The investigator studies the file's structure, metadata, readable strings, file type and executable characteristics.

Advantages

Static analysis can provide useful information while avoiding the risks associated with executing unknown malware.

Basic Techniques

- File identification
- Hash calculation
- String extraction
- Magic-byte analysis
- PE-header inspection



Strings Command

What is strings?

The **strings** command extracts sequences of readable characters from binary files.

A malicious executable may contain readable information such as:

- URLs
- IP addresses
- File paths
- Registry keys
- Error messages
- Command strings

Forensic Value

Strings can provide initial clues about what a suspicious binary may interact with.

Example

`strings suspicious.exe`

The output can then be reviewed for potentially interesting indicators.

```
fabio@fabio-S400CA:~$ strings file.exe
!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.rsrc
SSShL@X
E]u@8
QRP;6
7@JB
A/K?/?
/K7A?7/
JBCA
B@?/A
```



File Command and Magic Bytes

What is file?

The Linux **file** command identifies the type of a file by examining information within the file rather than relying only on its file extension.

Magic Bytes

Magic bytes are characteristic byte sequences associated with particular file formats. For example, executable files have identifiable structural characteristics.

Forensic Importance

This can help identify cases where:

- File extensions have been changed
- A file is disguised as another type
- The file type is unknown
- The extension does not match the actual content



PE-Bear and PE File Analysis

What is a PE File?

PE (Portable Executable) is the executable file format used by Windows for files such as .exe and .dll.

What is PE-Bear?

PE-Bear is a tool used to inspect the internal structure of PE files.

Areas of Examination

PE Headers

Provide structural information about the executable.

Sections

Show how different types of executable data are organised.

Imports

Show functions and libraries that the program may use.



PE analysis can help investigators understand the structure and potential capabilities of a suspicious Windows executable.



Integrated Network Investigation

Scenario

A workstation is suspected of communicating with a malicious server.

Investigation

Wireshark: Identify suspicious network traffic and follow TCP streams.



DNS Analysis: Identify unusual domain queries.



NetworkMiner: Extract hosts, files and other network artefacts.



Splunk: Correlate authentication and firewall logs.



IOC Identification: Record suspicious IPs, domains and file hashes.



Malware Analysis: Examine any recovered executable using static techniques.

Final Objective

Correlate **network + endpoint + malware evidence** to reconstruct the incident.



Thank you

